

Absolute Error and MAE

Project: Responsible Plant Classifier

1. Absolute Error and MAE

Teach the learning cycle

Primary target: Calculate MAE MSE and RMSE

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Look for: Fit mean and median constant baselines on training targets

Reference example, not the only acceptable wording: Teacher explanation: Mean 6; median 5.

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using evidence

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and prediction pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix $\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Concept to explain

Absolute error measures distance from truth without cancellation; MAE averages those distances in the target's original unit.

Term · Meaning

Term: absolute error

Meaning: absolute value of actual minus predicted

Term: MAE

Meaning: mean of absolute errors

Term: unit

Meaning: measurement scale retained by MAE

Term: cancellation

Meaning: positive and negative errors offset

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on absolute and squared loss. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

The training mean among all constant predictions.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain absolute error and mae using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Calculate residuals, absolute errors, and mean absolute error.

MODEL IDEA Absolute error measures distance from truth without cancellation; MAE averages those distances in the target's original unit.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual heights: 10, 14, 20; predictions: 12, 13, 17. Residuals: -2, +1, +3 cm; absolute errors: 2, 1, 3 cm.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Actual heights: 10, 14, 20; predictions: 12, 13, 17.

STEP 2 Residuals: -2, +1, +3 cm; absolute errors: 2, 1, 3 cm.

STEP 3 $MAE = (2 + 1 + 3) \div 3 = 2$ cm.

STEP 4 Compare with a baseline on the same cases and inspect the largest error.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Exact explanation to model

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Explain the added worked example

A loss summarizes error magnitude without positive and negative cancellation. MAE averages absolute residuals. MSE averages squared residuals and penalizes large magnitudes more strongly. RMSE takes the square root of MSE so it returns to the target unit. These are different questions, not interchangeable labels.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines absolute and squared loss.

Residuals [-1, 2, -3] cm have absolute errors [1, 2, 3], so $MAE = 6/3 = 2$ cm. Squared errors [1, 4, 9] sum to 14, so $MSE = 14/3$ square cm and $RMSE = \sqrt{14/3}$, about 2.16 cm. Keep the square and the average in the correct order.

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Key terms

Term · Meaning

Term: MAE

Meaning: Mean absolute error

Term: MSE

Meaning: Mean squared error

Term: RMSE

Meaning: Square root of mean squared error

Term: Loss

Meaning: A numerical penalty for prediction errors

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Absolute and squared loss at a glance

Part · Evidence or action · Interpretation

Part: Residuals

Evidence or action: minus 1 plus 2 minus 3

Interpretation: Signed differences

Part: Absolute values

Evidence or action: 1 2 3

Interpretation: MAE 2 cm

Part: Squares

Evidence or action: 1 4 9

Interpretation: MSE 14 divided by 3

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

Give MSE and RMSE with units. Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Absolute Error and MAE

Compare absolute and squared error without losing track of units.

The facts we will use

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/21

Absolute Error and MAE

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Absolute errors

Absolute values 1, 2, 3 give $MAE 6/3 = 2$ cm.



2. Squared errors

Squares 1, 4, 9 give $MSE 14/3$ cm², approximately = 4.67 cm².



3. Root squared error

$RMSE = \sqrt{14/3}$, approximately 2.16 cm.

CHANGE ONE THING

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Absolute errors

What happens and why: Absolute values 1, 2, 3 give $MAE 6/3 = 2$ cm.

Step: 2. Squared errors

What happens and why: Squares 1, 4, 9 give $MSE 14/3$ cm², approximately = 4.67 cm².

Step: 3. Root squared error

What happens and why: $RMSE = \sqrt{14/3}$, approximately 2.16 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. $RMSE = \sqrt{14/3}$, approximately 2.16 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Different loss functions weight large misses differently; a smaller selected loss is not automatically better for every consequence.

Change one thing in the pictured case

Change: Convert every residual to millimetres.

Prediction: MAE becomes 20 mm; MSE becomes $1400/3$ mm²; RMSE becomes about 21.60 mm.

Reason: Residuals scale by 10; squares scale by 100, while the square root returns a length unit.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Actual=[3,5,9], predicted=[2,7,6] cm.

Write the question to explain: Calculate MAE.

Reveal after predictions: Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Keep for an independent attempt: Actual heights are 4, 8 and 10 cm; predictions are 5, 6 and 13 cm. Calculate absolute errors and MAE.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: MSE $14/3$ square cm; RMSE $\sqrt{14/3}$ about 2.16 cm. Exit response: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Which constant minimizes training squared error? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify mae in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: compute mae; compute squared loss. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between mae and mse. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Calculate MAE MSE and RMSE; Explain how squaring changes sensitivity to large errors. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Give MSE and RMSE with units.

Teacher check: Teacher explanation: MSE $14/3$ square cm; RMSE $\sqrt{14/3}$ about 2.16 cm.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Absolute and squared loss. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: $(1+2+3)/3=2$ cm.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Its unit is cm^2 , not cm. RMSE is the corresponding square-root statistic with cm units. Changed-case check: MAE becomes 20 mm; MSE becomes $1400/3 \text{ mm}^2$; RMSE becomes about 21.60 mm. Residuals scale by 10; squares scale by 100, while the square root returns a length unit.

Changed-case reasoning: MAE becomes 20 mm; MSE becomes $1400/3 \text{ mm}^2$; RMSE becomes about 21.60 mm. Residuals scale by 10; squares scale by 100, while the square root returns a length unit.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. MAE Lab: calculate each row, verify units and denominator, then compare model and baseline on identical examples.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Average signed residual is a safe replacement for average absolute error.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: $(1+2+3)/3=2$ cm.

A stronger claim makes a stronger project: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: MAE 2 m; MSE 20/3 square m; RMSE about 2.58 m. In cm, MAE 200, MSE 200000/3, RMSE about 258.20.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Independent question

Actual heights are 4, 8 and 10 cm; predictions are 5, 6 and 13 cm. Calculate absolute errors and MAE.

Expected independent answer

Absolute errors are 1, 2 and 3 cm. MAE is $(1+2+3)/3 = 2$ cm. Signed errors must not cancel.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Averaging signed errors is the same as mean absolute error.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Residuals=[-2,3,-1], summing to 0. Absolute errors=[2,3,1], MAE=6/3=2 in the target's units. Signed errors can cancel even when every prediction is wrong.

Explanations for the additional workbook examples

W1 Compute MAE

Calculate MAE for the worked residuals.

Teacher explanation: $(1+2+3)/3=2$ cm.

MAE uses absolute residual sizes, so add 1, 2 and 3 and divide by three: $(1+2+3)/3 = 2$ cm. Positive and negative signed residuals must not cancel in this calculation. State centimetres because absolute error retains the target unit.

W2 Compute squared loss

Give MSE and RMSE with units.

Teacher explanation: MSE $14/3$ square cm; RMSE $\sqrt{14/3}$ about 2.16 cm.

Square the residual sizes to get 1, 4 and 9; their mean is $14/3$ square centimetres. Take the square root to obtain RMSE $\sqrt{14/3}$, approximately 2.16 cm. MSE and RMSE describe related quantities but have different units.

W3 Compare patterns

Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?

Teacher explanation: Both absolute sums are 6. Squared sums are 36 and 12, so MSE 12 versus 4.

W4 Apply independently

Residuals [-2, 0, 4] m: calculate MAE MSE RMSE and the result of converting to cm.

Teacher explanation: MAE 2 m; MSE $20/3$ square m; RMSE about 2.58 m. In cm, MAE 200, MSE $200000/3$, RMSE about 258.20.

W5 Choose honestly

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

Teacher explanation: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

Assess independent transfer

Selected case: Residuals [-2, 0, 4] m: calculate MAE MSE RMSE and the result of converting to cm.

Selected case answer

Teacher explanation: MAE 2 m; MSE $20/3$ square m; RMSE about 2.58 m. In cm, MAE 200, MSE $200000/3$, RMSE about 258.20.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 22: Train a Regression Model. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use mae and mse in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: MAE 2 m; MSE 20/3 square m; RMSE about 2.58 m. In cm, MAE 200, MSE 200000/3, RMSE about 258.20. Fresh retry residuals [-1, 1, 4]: MAE 2, MSE 6, RMSE sqrt 6 about 2.45 in corresponding units. Require units and operation order. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Fitting a line. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/22, ai-4/24, ai-4/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.